Discrimination in an Automated Society: Contributions from Latin America

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Do current Latin American laws protect people against discrimination due to the use of automated decision-making by artificial intelligence? How should an effective legal framework on this matter be configured? Based on these questions, this lecture analyzes this type of discrimination from the technical and regulatory point of view, provides a conceptual theory on the particularities of algorithmic discrimination situated and contextualized in Latin America, proposes the adoption of a relational principle based on a philosophical perspective, present in all Latin American indigenous peoples and offer a new interpretive approach to the principle of equality and the concept of discrimination under a collective view, in order to overcome normative challenges evidenced in comparative law.

Lecture Transcript

0:00 Hello. This lecture refer to algorithmic discrimination but from a beautiful and particular place, Latin America. I am Catherine Muñoz. I am a Chilean lawyer, I hold a Master's Degree in International Law, Investment Trade and Arbitration from the Universidad de Chile (University of Chile) and from Heidelberg University. I have always been fascinated in innovation and technology issues. So my career has followed those interests and I have been working at IP law firm for 14 years focused in IP and technology regulation and also I have been working as a consultant in public policy, ethical digital transformation, privacy and data governance. Since 2018, I have been working on regulatory issues related to the development and the use of AI systems. Also, I am currently part of the NGO OptIA made up by a group of outstanding interdisciplinary professionals. In English, the acronym OptIA means Observatory for Algorithmic Transparency and Inclusion for Latin America. At the same time, I have a personal project called Idonea AI, which is a consulting firm dedicated exclusively to legal, ethical and technological advice for the development of responsible digital technology and artificial intelligence.

2:01 There's a common pattern detected in the use of artificial intelligence systems, which assists or replaces the decision making processes that could normally be conducted by human when they're used in complex social issues. And such cases are directly excluded and oppressed groups such as African American, Latin, Indigenous peoples, LGBTIQ+ community, low income people, among others, are disproportionately affected, reproducing and perpetuating social injustices. Predicted risk model of of child's abuse that wrongly linked poverty with abuse predicted a crime system associating criminality when with a structural structural racism. Predicted system that detected welfare fraud are just a few examples. As written in the current legal framework, to regulate this type of AI is a priority in view of the exponential increase in the use of this technical solution in private sector and public institutions. So far different proposals has been developed mainly in Europe, Canada, some states of the US. This legal issue is extremely complex. And in opposition to intuition, this is not related to technological complexity, but rather to the legal such matter itself. Actually, equality and discrimination law has been built under historical and intergroup political and social tension with many different theories and interpretation. The concept of equality and discrimination have different contexts and scope in each country. Then, concept elaborated on the basis of their social context. Moreover, the antithesis, concept of discrimination in US, fairness has different interpretation depending on whether it is used in computer, social or legal sphere and with different implications of dependingon their social context. This look like the Tower of Babel.

4:47 Therefore it can be inferred why fairness doesn't have an exact equivalent in Latin America. It is no translatable term being interpreted has impartiality, equity or justice, which are not equivalent concept. Another complex concept is AI, artificial intelligence. I focus on the concept of AI for regulatory purposes. Regardless of the type of regulation, risk base or human right approach, referring to a legal definition of AI is inaccurate, since even though there are many definitions, none of them meet the appropriate legal parameters of proportionality, legal certainty, effectiveness and any attempt at definition could be restrictive. Therefore, it is desirable to consider a concept of AI for regulatory purposes, which should include the description of sociotechnical systems since, along with the technical component, the motivation of the people who are directly involved in the design and implementation of this type of technology is also important, as well as the particular social context in which these systems are envisage and develop. So, the sum of these technical and social factors explain the impact of using this system, especially the harmful impact. And this concept must be reflective, understanding that this technology can be used for different purposes and circumstances with different levels of risk. One of these impacts is algorithmic discrimination. So, perhaps you are wondering, how does AI really discriminate? That’s a very good question.

07:20 So, how algorithm discriminate? From a technical point of view, there is a traditional justification of bias in data and models that is the requisite for the generation of discriminatory act, which can be accidentally or intentionally generated. There are many types of bias in data. For example, one of the most well known is the bias arising through the inference biases embedded in the data itself. Indeed, data are rarely neutral since they are tied to people’s experience and history. Thus, reducing data within mathematical models without consideration of this rounded circumstance and context in order to provide a pretended neutrality inevitable leads to incomplete and misleading results. Another data bias, the representation bias, occur when only part of the population is sampled not representing the entire demographic group targeted by a particular system, usually under-representing or over-representing vulnerable groups in a negative way. Regarding bias in models, in machine learning models in particular, this can be explained by the designer’s choices and by their own technical limits. Machine learning are mathematical models with quantitative characters, so they perform data reduction not be able to analyze the variable founders' social construction. Behind this reduction, there is a history of development related to heuristic techniques widely established in technology and science, whose purpose is to reach a result quickly through circuits, which ultimately translate into common sense. The problem is that common sense, so using politics and power and discourses, is riddled with bias.

10:02 So, similarly, algorithms such as linear regression, were developed by 20th century eugenicists. And one of the most used algorithms in machine learning, hierarchical clustering, derives from our racist housing segregation model of the 1915 US. So, this algorithm have has a tendency or preference for the optimization of a predefined outcome and consequently generate exclusion has a counterpart. I took this historical reference from *Data*

Feminism, a book I recommend to everybody. It is also important to keep in mind that when we talk about prediction, we are referring to a result of correlation and not that predestination. Correlation as prediction are sensitive to changes in other contexts, and therefore, every bias. Nonetheless, explaining how discrimination is produced through a system exclusively from a technical perspective is a limitation, since, this is a sociotechnical concept that can only be explained very my purposes, motivation, social relationships that influence it, the balance and implementation. AI is situated in a context in which there has been an historical gap between men and women and exclusion of minority groups. So, it is designed and developed from a particular point of view of elite white men, because they are the one who hold the highest and most decisive position in the field of technological development.

12:24 Back to the legal sphere. The first approach regarding discrimination is related to establish what is meant by equality. Equality is the concept based on the Aristotelian notion of treating equals equally and unequals unequally. This basic formula present in international treaties, constitution or law now has equality before the law of formal equality. This definition needs to be complemented by external standards of justice to reveal a substantive character. The first interpretation of the standard justice indicates that equality is the guarantee of non discrimination. Discrimination can be carried when one person is treated less favorably that another person is, has been, or will be treated in a comparable situation on grounds such as gender, age, racism, religion, and others. In the US, this type of discrimination is called disparate treatment. Discrimination can also be indirect when there is apparently neutral practice, policy or rule which apply to everyone in the same way but put person who share a protected characteristic at disproportionate disadvantage compared with others. In US, this is called disparate impact. And this is the general rules of algorithmic discrimination. The challenges faced by the current non discrimination rules in a comporative law. Well, the first is finding a comparator. The comparator is an actual or hypothetical person in the same situation, who doesn't have the protector characteristic. In the case of algorithmic discrimination, finding a comparator is usually a complicated task, because in this case, the comparation are level of proof in very specific circumstances. Additionally, the concept of protected class or characteristic can be challenged. A new counter intuitive practice or marker of racism can also emerge that make it difficult to detect algorithmic discrimination. An additional challenge is that difficulties for the victim in proving a case and there is a lack of transparency. There are cases in which no detail of the system component may be necessary. For example, in the event that a particular data data set need to be analyzed to corroborate discriminatory biases in the data. In such circumstances, there may be legal barrier, such as trade secrets.

16:10 Now, there are additional challenges in Latin America. As I've already mentioned before, no discrimination law are situated in a particular context. And consequently, North American and European law adopt different approach and have different ways of assessing algorithmic discrimination since it is essentially social matter. From the Latin American context, even though we have a common history with our parent law, there is a legal nuance specific for each country that differ from the rest of the world. Likewise, from a technical point of view, in US research for example, a standard and metrics for bias detection are focused on racism against black people. This bias detection may not be entirely applicable for other regions, such as Latin America when race's markers are very different. For example, a mechanism to the tech discrimination, considering a skin tone as a marker of racism, hypothetically may not be able to detect discrimination in Latin America, where skin tone is composed of shade. This means that not only it will be no useful, but equally generate a false negative result of discrimination, which is why the development of the regional, technical and legislative study on this issue in Latin America is very important. So, which are the specific Latin American elements to be consider in algorithmic discrimination issue? First, the history of Latin America from its origins must be taken into consideration. From this historical perspective, it is possible to realize that race as a social construct and class, also gender, are deep deeply intertwined may be difficult to detect or differentiate one category from other in for example, an algorithmic bias assessment. Disconnection of of protected category has emerged from two historical event that occurs simultaneously - the constitution of America and the creation of colonial capitalism. So, the social classification of the colonized population or the idea of race serve in the first place, as has a justification for the new European elimination, a practice called colonialism. This classification leads to new social identities - Indian, Africa, African, Mestiza and redefine concepts such as Spanish or Portuguese, from just simple geographical indicator to hierarchical sign of humans as priority. This classification by phenotypical features serve at the same time for the cultural exploitation of labor in place to the development of capitalism and the class, the social class. Therefore, the class and the concept of race were a structurally from their origin until today. Discrimination generate of the by the basis of this category that is racism, class and gender are etc that is is the result of concatenated historical circumstances, no big in any case independent or sporadic event.

20:38 A further characteristic of the region, especially in the South Pacific Area, is that country have a majority of mixed race population, with a low proportion of European and indigenous people, but the original racist hierarchical structure has remained in place to this day. So, in countries such as Mexico, Colombia, Argentina, Chile and Uruguay, there is a deep rooted, rooted belief about overcoming racial problems based on the existence of generalized message, however, as mentioned before in these societies, hierarchical structure based on their own racism has been maintained. This dissonance called racial color blindness is characterized by the fact that people believe that inequality, exclusion and discrimination are not racism same has a problem of the past that has been overcome, explaining any type of injustice from other angles such as the consequence of the market, nature or luck. Racial color blindness hinders the detection of algorithmic discrimination, since it is probable that a person may not be able to perceive has been discriminated for on racial grounds and may attribute some discriminatory situation so to unfair act, or nature or random events. Likewise, there is a tendency in some Latin America countries to classify themselves as white people. So in conclusion, discrimination arising from the use of AI system in Latin America has a particularity specific to the region. First, the suspicious characteristics of race and class, as well as gender, are unified by complex intersectional and structural layers. Thus, in addition to hiding racism as a relevant cultural and social problem, it can be difficult to determine which category is being affected in, for example, an algorithmic bias analysis. The racial color-blindness make difficult for people to be aware by themselves, that they are being discriminated, which is exacerbated by the fact that the same color blindness allow many practice which are clearly discriminatory acts as socially accepted or tolerated.

24:07 Proxies or markers and metrics for detecting discriminatory bias in AI system which has been developed in the Global North are neither compatible nor transferable to Latin American reality. Here it is possible to distinguish different and unique proxies for racism, such as subtle skin tone, maiden name and last name as a whole with the same importance. Phenotypic aspects as height, hair, eyes and nose shapes, they're also proxies for classism. As zip code, primary or secondary school, educational levels, first generation college student, person's neighborhood, district, vocabulary, idioms, clothing and more many others. What is the problem? The current technical and legislative research and proposals on methods and parameters for detecting algorithmic biases cannot be transferred to the Latin America context, which is why the development of regional research on this issue is so important. So, what is needed? Regulatory framework as well as metric and standard of bias detection development for Latin America based on praxis and social context specific for the region. As a starting point I suggest a normative conceptualization composes of three elements: the adoption of relational principle based on a philosophical perspective present in all Latin American indigenous people, a new interpretive approach to the principle of equality and a reinterpretation of the concept of discrimination under a collective view in order to overcome normative challenges evidenced in comparative law.

26:43 The first element is the relational principle the indigenous people in Latin America have in common - our relational worldview. This includes a community experience, contrary to any concept of individuality, and competition. Despite differences in language and culture, this transversal principle common in all country of the region, is also shared with indigenous people of Africa. There are some dissections of family working or living in community with solidarity and mutual help characterize all indigenous people of the region. In view of this, it is important for a rescue this principle that can serve as a source to reinterpreted the principle of equality and the concept of discrimination. Also be a basis for a new positive profitable relationship with this technology.

28:00 The second element equality as antisubordination from a collective perspective. The Inter American Commission on Human Rights has recently stated that human rights are not a static concept and must be constantly reinterpreted in the light of social change. In this regard, it will be noted that equality must also be interpreted from a collective perspective following Iris Marion Young considering equality has an individual principal whose evaluation is based on individuals is a wrong approach. Therefore, in order toconclude that something is unjust or non discriminatory or not, the evaluation must necessarily be made in terms of groups. From this collective approach, it is possible to know that any social, racial and gender inequality are a structural problem rather than merely a pity or irrational sporadic act. Moreover, classical equality, understood has no discrimination guarantee has a neutral character, doesn't touch the extractor of social injustice and maintains the status quo. Bearing in mind our Latin America historical context, it is necessary to interpret the principle of equality not only from a collective perspective, but as an anti-subordination principle. This means that the purpose of the principle of equality goes beyond a guarantee against discrimination. Its purpose being to dismantle social structures that discriminate and exclude certain groups. It is more active. This perspective of equality is significant at the moment of evaluating the person of discrimination through the use of algorithimicism. And since it's clarified that the fact that any unequal treatment of protected groups, whether intention, intentional or not, is not tolerated and must be condemned, since its result reproduce and perpetuate the structural condition of social injustice. Discrimination must be also be understood from a collective point of view, since when someone suffered discrimination besides that violation of their their human right, a category shared by a group of people is impacted. Therefore, a whole group of the same directly affected through the amplification and perpetration of the structural discrimination. Likewise, is relevant a notion of discrimination that can deal with new paradigms, such as algorithmic discrimination with different social contexts. For example, in this regard, Issa Kohler Hausmann, has defined as discrimination has an action or practice that act or reproduce an aspect of a category in a morally objectionable way. This concept has some complex ethical content. In order to establish when an act has been discriminatory, it must be started from a moral norm continually constituted explanation that describe how an action or practice is morally objectionable taking into consideration the complex connection of meaning and social relation that constitute one category and is structural content. Rather than the classic definition of discrimination that use competitor and mathematical metrics. Thus, apprehension about how legislation might respond to new groups, or new wave of discrimination against certain groups, could be covered by this reflexive hypothesis. This theory is critical from the classic way of the of the detecting discrimination, the counterfactual causal model, which focus on our evaluation of a protected characteristic, isolating it from its context. For example, according to a recent investigation in US, it was determined that the percentage of white and black people arrested by police is similar. That is, there is no racial bias in this aspect. But behind this statistic, there is a whole empirical evidence and historical racism that make it difficult to believe this balance and that can be explained through the search of this more morally objectionable act. In doing so, we can consider other additional facts, such as the fact that the data itself is biased. Indeed, it has been documented from multiple, multiple cases, that the police intentionally attempt to maintain in their records the same number of arrests, say number of white people and black people deliberately. For example, stop by people for artificial or minor reason without any consequence to them. This important when determining if discrimination occurred through algorithm, since method for detecting fairness has mathematical methods or comparator can not can be reflect this social construction. Therefore, this method must necessarily be complemented by other qualitative methodology.

